

WAJAHAT AHMED

AI ENGINEER — MACHINE LEARNING, LLMs & COMPUTER VISION

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PROFESSIONAL SUMMARY

AI Engineer with 4 years of experience building and deploying production AI systems across LLM/RAG applications, computer vision, and edge AI. Skilled in designing scalable ML pipelines and AI microservices RAG systems, LLM fine-tuning, and real-time inference. Experienced in taking models from research to production with CI/CD, containerization, and continuous performance monitoring.

CORE SKILLS

- Programming: Python, C++, C, MATLAB, SQL
- Machine Learning: PyTorch, TensorFlow, Scikit-learn, Deep Learning, Feature Engineering, Model Evaluation, Time-Series Analysis
- Computer Vision: OpenCV, YOLO, MMDetection, MMPose, MediaPipe, CNNs (2D/3D), Object Detection, Pose Estimation, Multi-Object Tracking, Image Classification, Video Analytics
- Generative AI & NLP: LLMs, SLMs, RAG Pipelines, LangChain, LangGraph, LLM Fine-Tuning (LoRA/QLoRA/PEFT), Vector Databases (Milvus, Qdrant)
- Data & MLOps: FastAPI, Docker, Git, Linux, CI/CD, Model Monitoring, Pandas, NumPy, PostgreSQL, MongoDB, Milvus, Redis, GCP
- Embedded & Edge AI: Jetson Nano, Raspberry Pi, ESP32
- Tools & Collaboration : Jira, ClickUp, Notion, Slack, Agile

EXPERIENCE

AI Developer — Lambda Theta | Sep 2024 – Present

- Led the end-to-end development and delivery of AI solutions for enterprise clients, owning project architecture, technical decisions, deployment, and customer delivery from requirements gathering to production rollout.
- Owned the complete lifecycle of multiple AI features, from solution design and implementation to deployment, maintenance, and customer support, serving as the primary technical owner throughout development.
- Designed and deployed scalable computer vision systems for real-time video analytics across distributed multi-camera environments, processing 20+ concurrent RTSP streams while maintaining low-latency inference and reliable performance.
- Built end-to-end machine learning pipelines covering data collection, annotation, preprocessing, model training, evaluation, deployment, and continuous performance monitoring on datasets exceeding 200K+ labeled samples and 1,200+ video sequences.
- Designed hybrid deep learning architectures combining spatial and temporal representations to improve recognition of complex human activities under viewpoint variation, occlusion, and concurrent actions.
- Investigated model failure modes, distribution shifts, and deployment challenges, applying geometric and statistical approaches to improve model robustness and real-world generalization.
- Evaluated and optimized machine learning models—including Random Forest, XGBoost, LightGBM, MLP, RNN, GRU, and LSTM—to identify architectures balancing accuracy, latency, and computational efficiency for production systems.
- Fine-tuned and deployed computer vision models using PyTorch, YOLO-Pose, MediaPipe, and MMDetection, including solutions for human action recognition, drink spiking detection, and satellite imagery analysis.

- Developed Retrieval-Augmented Generation (RAG) systems and AI agent workflows using embedding models, vector databases, LangChain, LangGraph, and LLM APIs for semantic search and autonomous multi-step reasoning.
- Architected and developed production-ready AI microservices using FastAPI, integrating PostgreSQL, Redis, and Milvus for scalable document ingestion, embedding generation, vector indexing, and semantic retrieval.
- Containerized AI services using Docker, integrated CI/CD pipelines for automated deployment, and supported scalable deployments across local GPU infrastructure and Google Cloud Platform (GCP).
- Collaborated directly with customers to define technical requirements, translate business challenges into AI solutions, and deliver production-ready systems with measurable outcomes.
- Authored technical documentation, architecture designs, and project proposals while mentoring junior engineers and interns, providing technical guidance throughout project execution.

Research Assistant — Neurocomputation Lab, National Center of AI | Sep 2022 – Sep 2024

- Designed and deployed AIoT-enabled systems for intelligent energy monitoring and optimization, integrating embedded hardware, IoT communication, real-time data acquisition, and machine learning to support practical energy-efficient solutions.
- Developed end-to-end machine learning workflows—including data preprocessing, feature engineering, model training, evaluation, and statistical analysis—for large-scale sensor datasets collected from real-world deployments.
- Evaluated and optimized machine learning models including Random Forest, Decision Tree, SVM, KNN, ANN, LSTM, and Hidden Markov Models to predict user-preferred fan speed while minimizing energy consumption.
- Designed and developed AIoT-enabled embedded hardware and firmware for intelligent ceiling fan automation, enabling continuous sensing, remote monitoring, and long-term field deployment.
- Designed experimental platforms for Li-ion battery characterization, integrating automated sensing, embedded controllers, and data acquisition pipelines to generate datasets for battery health prediction and performance analysis.
- Developed low-power embedded firmware for ESP32 and edge devices, integrating BLE communication and heterogeneous sensor networks for reliable real-time environmental and operational data acquisition.
- Deployed deep learning models on Raspberry Pi and embedded edge platforms for real-time computer vision applications under latency and computational constraints.
- Conducted firmware security assessments for embedded controllers used in electric vehicle systems, identifying vulnerabilities and implementing mitigation strategies to improve system reliability.
- Designed, integrated, and validated intelligent cyber-physical systems, managing hardware integration, sensor calibration, field testing, and performance evaluation across multiple research projects.
- Supervised undergraduate research projects in AI, embedded systems, and robotics while mentoring students on system design, experimentation, and machine learning methodologies.
- Authored and co-authored peer-reviewed publications, research proposals, technical reports, and project documentation communicating experimental methodologies, engineering designs, and research outcomes.

Selected Projects

- **Qiyas – Distributed Multi-Camera Activity Recognition:** Developed a scalable AI-based perception system for distributed multi-camera human activity recognition using hybrid spatial-temporal deep learning models. Designed for real-time deployment across surveillance networks, achieving 95% detection accuracy under real-world conditions.

- **InstruX – Industrial Energy Intelligence Platform:** Developed an industrial AIoT platform integrating multiple energy analyzers for real-time monitoring, visualization, and predictive analytics. Designed machine learning models for energy forecasting and optimization, resulting in a journal paper currently under review.
- **Hazard Detection:** Developed a 3D CNN-based fire detection system for real-time CCTV surveillance, optimizing spatio-temporal feature extraction for early hazard recognition.
- **Document Intelligence Platform:** Architected a production-ready document intelligence platform supporting document ingestion, embedding generation, vector search, and semantic retrieval through modular AI microservices built with FastAPI and modern vector databases
- **SmartFan – AIoT Smart Fan System:** Designed an AIoT-enabled smart fan system integrating embedded hardware, IoT communication, and machine learning for intelligent energy optimization and user comfort. Co-authored a peer-reviewed publication presenting the system architecture and experimental validation.

EDUCATION

- M.S. Artificial Intelligence — NED University of Engineering & Technology (2026 – Present)
- B.S. Applied Physics (Electronics Major, Gold Medalist) — NED University of Engineering & Technology (2018 – 2022)

PUBLICATION

Hashim Raza Khan, Wajahat Ahmed et al., “Design and Experimental Results of an AIoT-Enabled, Energy Efficient Ceiling Fan System,” MDPI Sustainability, 2024.